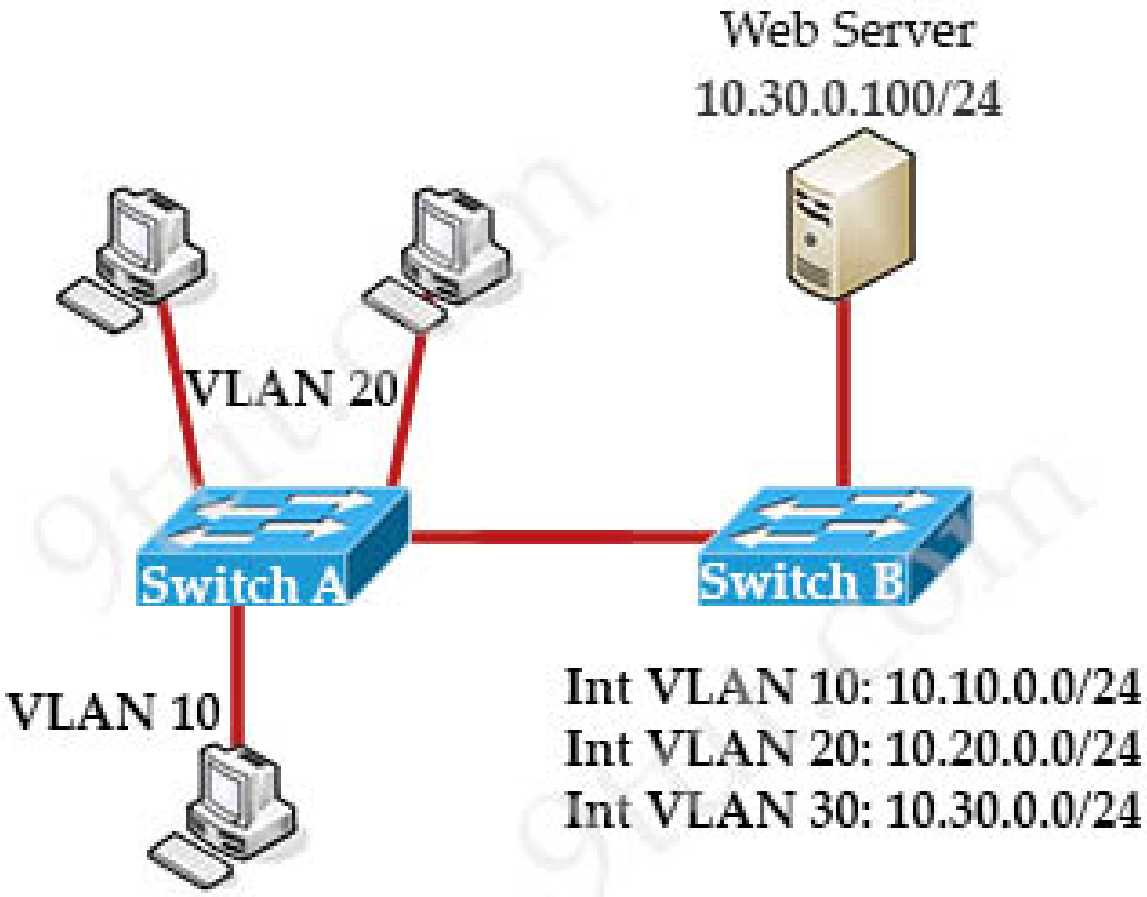
Access-list Questions

7 questions | Original English questions and source answers

Question 1

Refer to the exhibit. A network engineer must block access for all computers on VLAN 20 to the web server via HTTP. All other computers must be able to access the web server. Which configuration when applied to switch A accomplishes this task?



Options

A.
```
onfig t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
int vlan 20
ip access-group wwwblock in
```

B.
```
config t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
int vlan 100
ip access-group wwwblock in
```

C.
```
config t
ip access-list extended wwwblock
permit ip any any
deny tcp any host 10.30.0.100 eq 80
int vlan 30
ip access-group wwwblock in
```

```
D. config t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
permit ip any any
int vlan 20
ip access-group wwwblock in
```

Answer

D. config t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
permit ip any any
int vlan 20
ip access-group wwwblock in

Source explanation

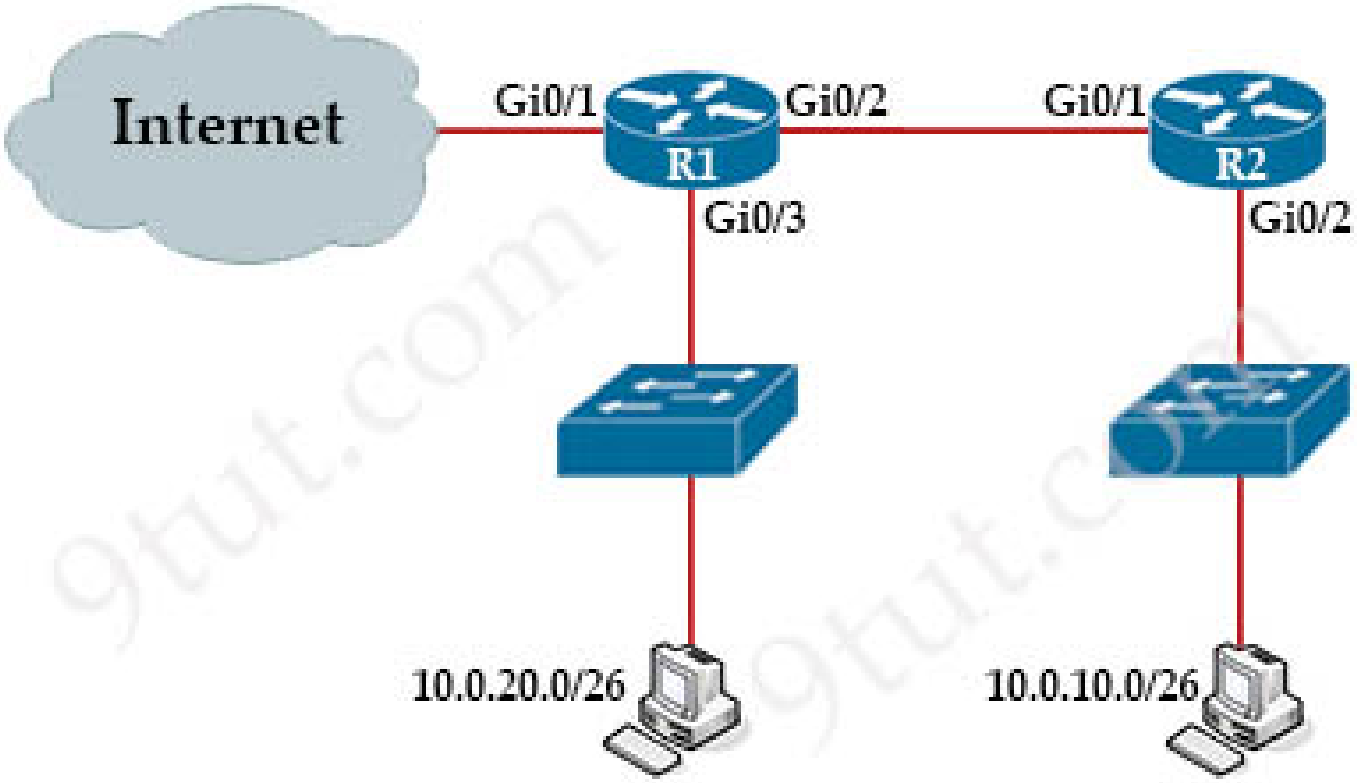
The “deny tcp any host 10.30.0.100 eq 80” command means “block all (any) traffic from accessing web server at 10.30.0.100 on port 80”. And since it is applied to VLAN 20 interface so only computers on VLAN 20 are affected.
In summary, just notice that 10.30.0.100 here is the destination IP address, not source address.
Note: The traffic flow from hosts in VLAN 20 to the Web Server is: host in VLAN 20 -> Interface VLAN 20 -> Interface VLAN 30 -> Web Server. If we place the ACL: host in VLAN 20 -> (ACL Inbound) Interface VLAN 20 -> Interface VLAN 30 -> Web Server. Therefore the ACL can block traffic from VLAN 20.

Source: CCNA Training » Access-list Quiz.pdf | Question 1

Question 2

Refer to the exhibit. An extended ACL has been configured and applied to router R2. The configuration failed to
work as intended. Which two changes stop outbound traffic on TCP ports 25 and 80 to 10.0.20.0/26 from the 10.0.10.0/26 subnet while still allowing all other traffic? (Choose two)

```
R2#config t
R2(config)#access-list 101 deny tcp 10.0.20.0 0.0.0.63 10.0.10.0 0.0.0.63 eq smtp
R2(config)#access-list 101 deny tcp 10.0.20.0 0.0.0.63 10.0.10.0 0.0.0.63 eq www
R2(config)#int gi0/2
R2(config-if)#ip access-group 101 in
```



Options

A. The source and destination IPs must be swapped in ACL 101

B. Add a “permit ip any any” statement at the end of ACL 101 for allowed traffic

C. The ACL must be configured the Gi0/2 interface inbound on R1

D. The ACL must be moved to the Gi0/1 interface outbound on R2

E. Add a “permit ip any any” statement to the beginning of ACL 101 for allowed traffic

Answer

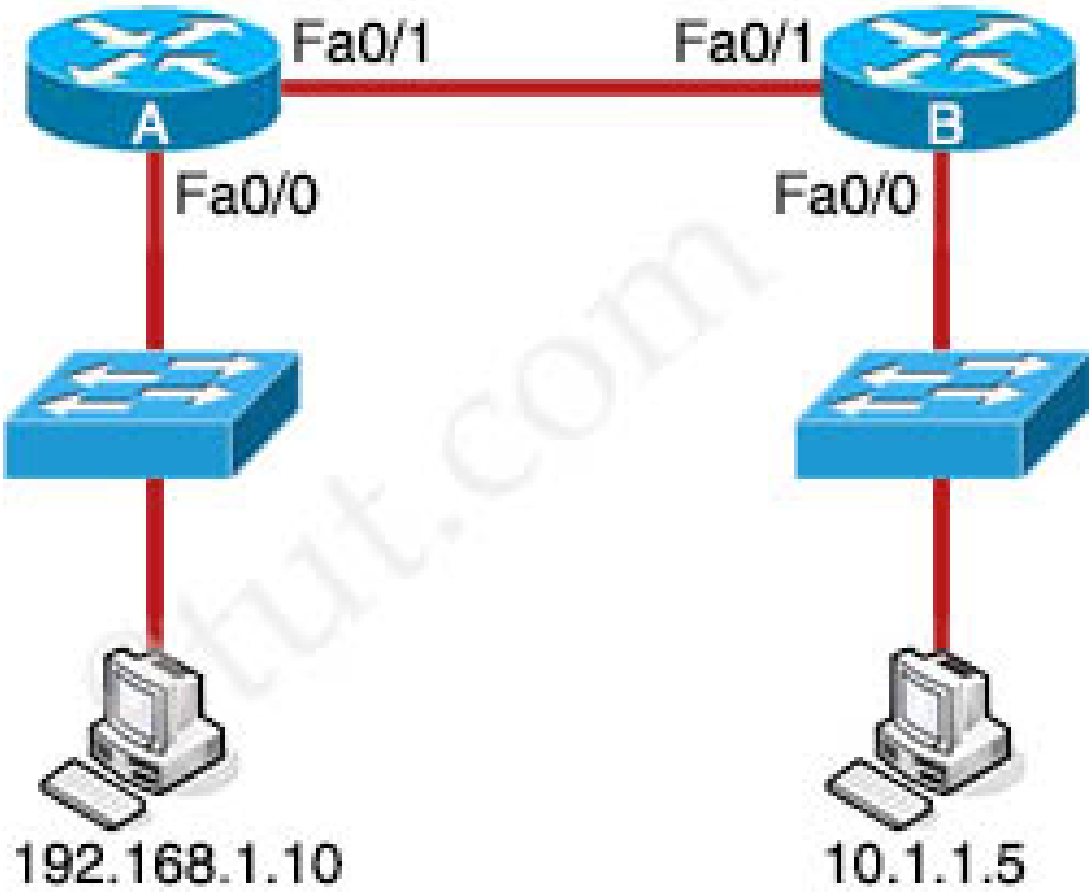
A. The source and destination IPs must be swapped in ACL 101

B. Add a “permit ip any any” statement at the end of ACL 101 for allowed traffic

Source: CCNA Training » Access-list Quiz.pdf | Question 2

Question 3

Refer to the exhibit. An administrator configures the following ACL in order to prevent devices on the 192.168.1.0 subnet from accessing the server at 10.1.1.5:
access-list 100 deny ip 192.168.1.0 0.0.0.255 host 10.1.1.5
access-list 100 permit ip any any
Where should the administrator place this ACL for the most efficient use of network resources?



Options

A. inbound on router A Fa0/0

B. outbound on router B Fa0/0

C. inbound on router B Fa0/1

D. outbound on router A Fa0/1

Answer

A. inbound on router A Fa0/0

Source explanation

By placing the ACL closest to the source, we can reduce unnecessary traffic passing between two routers and the processing time of the router.

Source: CCNA Training » Access-list Quiz.pdf | Question 3

Question 4

Refer to the exhibit.
interface GigabitEthernet0/1
ip address 192.168.1.2 255.255.255.0
ip access-group 2699 in
!
access-list 2699 deny icmp any 10.10.1.0 0.0.0.255 echo
access-list 2699 deny ip any 10.20.1.0 0.0.0.255
access-list 2699 permit ip any 10.10.1.0 0.0.0.255
access-list 2699 permit tcp any 10.20.1.0 0.0.0.127 eq 22
A network administrator must permit SSH access to remotely manage routers in a network. The operations team
resides on the 10.20.1.0/25 network. Which command will accomplish this task?

Options

A. `access-list 2699 permit tcp any 10.20.1.0 0.0.0.255 eq 22`

B. `access-list 2699 permit udp 10.20.1.0 0.0.0.255`

C. `no access-list 2699 deny ip any 10.20.1.0 0.0.0.255`

D. `no access-list 2699 deny tcp any 10.20.1.0 0.0.0.127 eq 22`

Answer

C. no access-list 2699 deny ip any 10.20.1.0 0.0.0.255

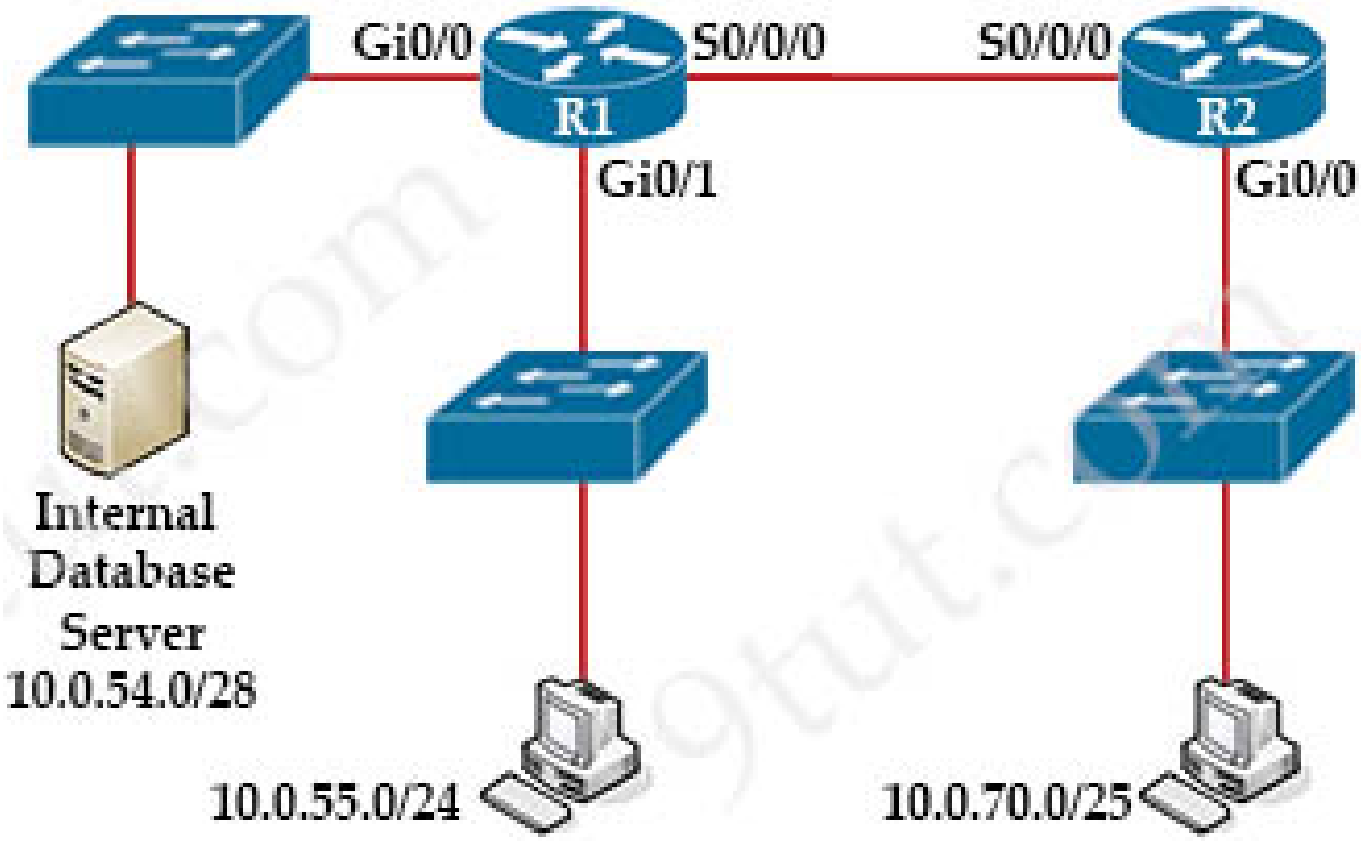
Source explanation

The operations team resides on 10.20.1.0/25 network which is a part of 10.20.1.0/24 network so we need to remove the “deny” statement of the 10.20.1.0/25 network to allow SSH.
Note: the ACL is blocking the reply traffic so 10.20.1.0/25 should be the destination.

Source: CCNA Training » Access-list Quiz.pdf | Question 4

Question 5

Refer to the exhibit. Which two configurations would be used to create and apply a standard access list on R1, so that only the 10.0.70.0/25 network devices are allowed to access the internal database server? (Choose two)



Options

A. R1(config)# access-list 5 permit any

B. R1(config)# access-list 5 permit 10.0.54.0 0.0.1.255

C. R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip access-group 5 out

D. R1(config)# interface Serial0/0/0
R1(config-if)# ip access-group 5 in

E. R1(config)# access-list 5 permit 10.0.70.0 0.0.0.127

Answer

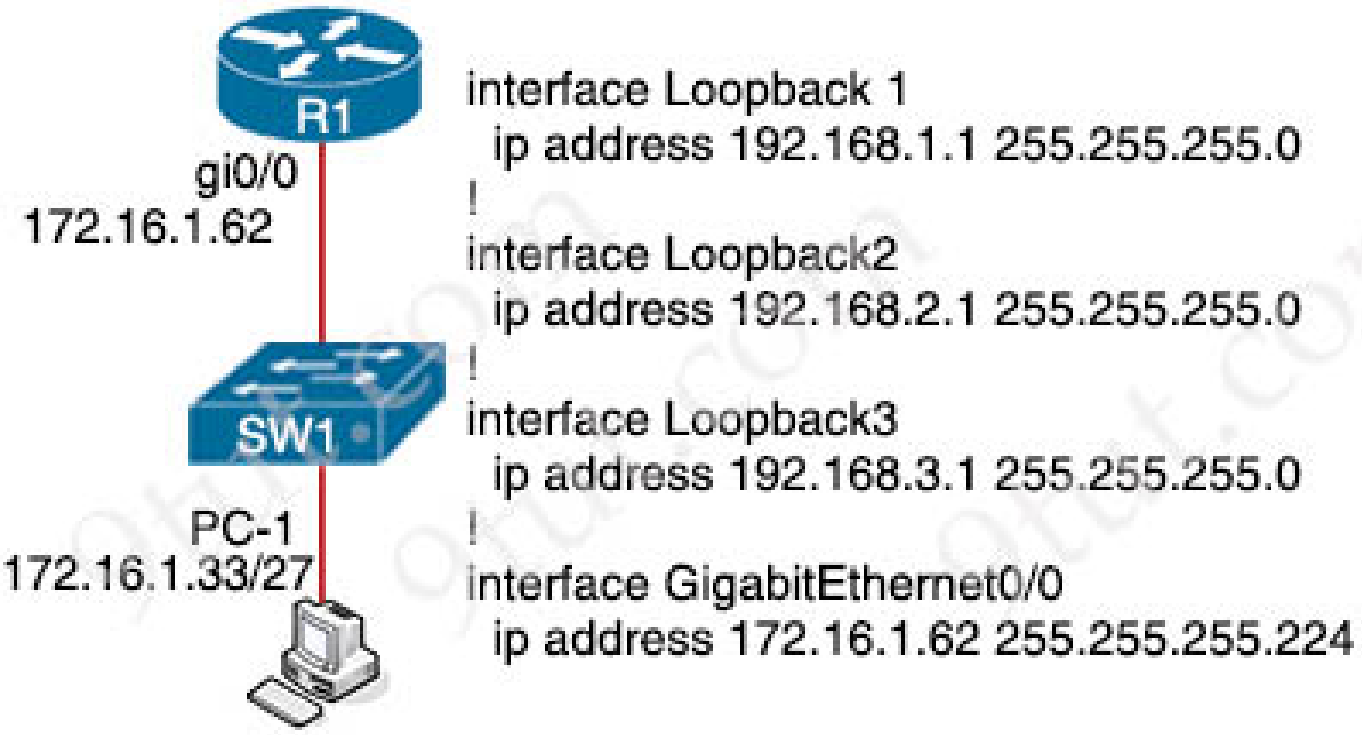
C. R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip access-group 5 out

E. R1(config)# access-list 5 permit 10.0.70.0 0.0.0.127

Source: CCNA Training » Access-list Quiz.pdf | Question 5

Question 6

Refer to the exhibit. What configuration on R1 denies SSH access from PC-1 to any R1 interface and allows all other traffic?



Options

A.
```
access-list 100 deny tcp host 172.16.1.33 any eq 22
access-list 100 permit ip any any
```

B.
```
access-list 100 deny tcp host 172.16.1.33 any eq 22
access-list 100 permit ip any any
interface GigabitEthernet0/0
ip access-group 100 in
```

C.
```
line vty 0 15
access-class 100 in
access-list 100 deny tcp host 172.16.1.33 any eq 23
access-list 100 permit ip any any
interface GigabitEthernet0/0
ip access-group 100 in
```

D.
```
access-list 100 deny tcp host 172.16.1.33 any eq 23
access-list 100 permit ip any any
line vty 0 15
access-class 100 in
```

Answer

B. access-list 100 deny tcp host 172.16.1.33 any eq 22
access-list 100 permit ip any any
interface GigabitEthernet0/0
ip access-group 100 in

Source explanation

To deny SSH, we must block TCP port 22 and we need to apply the ACL to Gi0/0 interface (which connects to PC-1).
Note: TCP port 23 is used for Telnet.

Source: CCNA Training » Access-list Quiz.pdf | Question 6

Question 7

While examining excessive traffic on the network, it is noted that all incoming packets on an interface appear to
be allowed even though an IPv4 ACL is applied to the interface. Which two misconfigurations cause this behavior? (Choose two)

Options

A. A matching deny statement is too high in the access list

B. A matching permit statement is too high in the access list

C. A matching permit statement is too broadly defined

D. The ACL is empty

E. The packets fail to match any permit statement

Answer

B. A matching permit statement is too high in the access list

D. The ACL is empty

Source explanation

If we have a matching permit statement too high in the access list then it will be matched first before we can deny i -> Answer B is correct.
The statement that an ACL always has an implicit deny any at the bottom has one exception. And that exception
is when the ACL is empty. If you use ip access-group to apply an ACL and that ACL has no statements then all traffic is permitted -> Answer D is correct.
Reference: https://community.cisco.com/t5/routing/apply-empty-acl-what-happens/td-p/740473
Note: Answer "A matching permit statement is too broadly defined" seems to be correct too but we only have two choices in this question and this answer is not in the best answers. If a matching permit statement is too broadly defined (for example if we only want to permit TCP traffic then we should not permit “ip” traffic, which includes both TCP and UDP).

Source: CCNA Training » Access-list Quiz.pdf | Question 7